



Semester I Examinations 2016/2017

Exam Codes	3BA1, 3BS9, 3MR3, 3BPT1, 4BA4, 4BME1, 4BS2, 1CNS1 1OA1, 1EM1
Exams	Third Arts, Third Science, Third Marine Science, Fourth Arts, Fourth Mathematics and Education, Fourth Science, Master of Science
Module	Applied Statistics I
Module Codes	ST311
External Examiner	Professor S. Wilson
Internal Examiner(s)	Dr. E. Holian Professor J. Hinde
<u>Instructions:</u>	Answer any 3 questions from Questions 1-4 All Questions carry equal marks.
Duration	2 Hours
No. of Pages	22 pages, including this one
School	School of Mathematics, Statistics and Applied Mathematics
Release to Library:	Yes ☒ No ☐
Release in Exam Venue	Yes ☒ No ☐
<u>Requirements:</u>	
Statistical Tables/ Log Tables	Relevant distribution tables are attached to this paper. The <i>New Formulae & Tables</i> are optional.
Other Materials Allowed	A calculator is allowed (non-programmable and not capable of storing text).

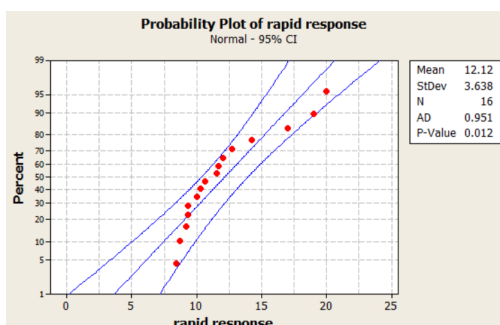
Question 1

Complete parts (a), (b) and (c).

- (a) An emergency rapid response team were observed carrying out repeated trials of a standard training exercise. The recorded times to complete the exercise for the sample of 16 trials observed for the response team, in minutes, are:
8.4, 8.7, 9.2, 9.3, 9.3, 10, 10.3, 10.6, 11.5, 11.7, 12, 12.7, 14.2, 17, 19, 20.
A summary of descriptive statistics and plots for this sample are provided below:

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3
rapid response	16	0	12.119	0.910	3.638	8.400	9.300	11.050	13.825

Variable	Maximum	Mode	N for Mode	Skewness	Kurtosis
rapid response	20.000	9.3	2	1.20	0.41



- By comparing the values of the mean, the median and the mode(s) for this sample what does this tell us about the shape of the distribution of the sample? [2 marks]
 - What does the normal probability plot suggest? [2 marks]
 - Construct the letter value display for this sample, including midvalues and spreads. [4 marks]
 - What do the letter values indicate about the sample distribution? [4 marks]
- (b) The Chief in charge of the rapid response team wishes to determine whether this sample gives evidence that the true typical time to complete the exercise is greater than 10 minutes.
- It is suggested that a 1-sample t-test be carried out. The output is included below. Assuming the assumptions of the test hold what can you conclude as a result of this analysis? Reference the output to justify your answer. [3 marks]

One-Sample T: rapid response

Test of $\mu = 10$ vs > 10

Variable	N	Mean	StDev	SE Mean	95% Lower Bound	T	P
rapid response	16	12.119	3.638	0.910	10.524	2.33	0.017

Question 1 continues on the next page

Question 1 continued...

- (ii) Do you have any concerns about applying the 1-sample t-test in this application? If so, explain reasons for your concern. [2 marks]
- (iii) It is suggested that a 1-sample Sign test be carried out. Use the sample to test where there is evidence to suggest $\eta > 10$, testing at $\alpha = 0.05$. Include a statement of the hypotheses being tested, an explanation of how the p-value is calculated defining any symbols you use, your decision and conclusion in this application. [7 marks]
- (c) A company is launching a new brand of emergency flare to the market and have been asked to demonstrate that this brand performs better in terms of burning time than the leading competitor brand of emergency flare.

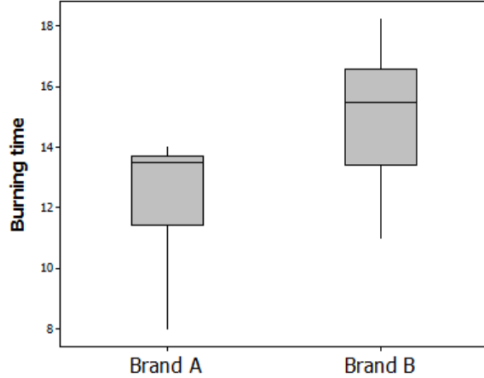
The company obtained a sample of 8 flares of the leading competitor flare, Brand A, and a sample of 10 flares of their new brand, Brand B. The burning times observed for these samples are:

Brand A: 8, 11.2, 12, 13.4, 13.6, 13.6, 13.7, 14.

Brand B: 11, 11.6, 14, 14.5, 15.1, 15.8, 16.1, 16.3, 17.3, 18.2.

Some descriptive statistics and plots for these samples are also provided.

Variable	N	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Brand A	8	12.44	0.721	2.038	8.00	11.40	13.50	13.68	14.00
Brand B	10	14.99	0.729	2.307	11.00	13.40	15.45	16.55	18.20



The researcher wishes to carry out a hypothesis test to infer whether the samples give evidence that the typical value of burning times for the population of Brand B flares is greater the typical value of burning times for the population of Brand A flares.

- (i) It is suggested to you to carry out a Wilcoxon Rank Sum test. Explain why, in this application, this test is more suitable than the 2-sample Z test, or the 2-sample t test. [4 marks]
- (ii) Carry out the test at significance level $\alpha=0.05$. Include a statement of the hypotheses being tested, formulation of the test statistic, the rejection region for the test statistic, and indicate clearly your decision and conclusion from the result of the test. [7 marks]

[Total 35 marks]

Question 2

Complete parts (a), (b) and (c).

- (a) (i) "The true correlation coefficient between two variables X and Y for a population of individuals is known to have value $\rho = 0$. This implies that X and Y are unrelated."
- Is this statement true or false? If you deem the statement to be false give a reason for your answer. [2 marks]
- (ii) A sample of 15 individuals gave a sample correlation coefficient of $r = 0.59$. Carry out a hypothesis test of whether there is evidence of a significant increasing relationship between X and Y in the population. Include clear statements of the hypotheses, the rejection region and conclusion. [6 marks]
- (b) Consider a simple linear regression model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$.
- (i) What does the error term, ϵ_i , represent? [2 marks]
- (ii) When carrying out an Analysis of Variance, explain in general what the terms SS_{Total} , SS_{Reg} and SS_{Error} represent. [5 marks]
- (c) The following model is being considered in order to model the variability in a response variable Y , using four candidate predictor variables, X_1, X_2, X_3, X_4 ,
- $$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \epsilon_i.$$
- The model is fitted to observations obtained for a sample of 15 individuals, and the results of the fit are provided in the partial printout on the next page. Use the output to answer the following questions.
- (i) Calculate the coefficient of determination R^2 for this regression fit. Interpret the meaning of this value. [2 marks]
- (ii) Find the missing values in the ANOVA table, and complete the test, at the $\alpha = 0.05$ level, including statements of the hypotheses being tested, a rejection region, a decision and conclusion. (*You do not have to calculate the p-value.*) [6 marks]
- (iii) Complete the missing t-test in the output, i.e. test whether X_3 is a good predictor of Y at the $\alpha = 0.05$ level. Include the hypotheses being tested, the calculation of the test statistic, a rejection region, a decision and conclusion. [5 marks]
- (iv) The output references an individual in the sample that has a large standardised residual. This observation has predictor values $X_1 = 12$, $X_2 = 6$, $X_3 = 24$, $X_4 = 70$, and observed response $Y = 245.95$. Find the value of the fitted response $\hat{Y}$ and the fitted residual $\hat{\epsilon}$ for this individual, and thus show how the standardised residual of 244 is arrived at. Why has this observation been flagged as an unusual observation? [5 marks]
- (v) The output provides a prediction interval and a confidence interval for the response when $X_1 = 10$, $X_2 = 12$, $X_3 = 25$, $X_4 = 60$. Explain the difference as to when a prediction interval and a confidence interval applies. [2 marks]

Question 2 continues on the next page

Question 2 continued...

The regression equation is

$$Y = 80.5 + 0.71 X_1 - 2.93 X_2 + 4.26 X_3 - 0.10 X_4$$

Predictor	Coef	SE Coef	T	P
Constant	80.50	66.38	1.21	0.253
X1	0.71	1.463	0.49	0.636
X2	-2.93	1.020	-2.87	0.017
X3	4.26	1.794	-	-
X4	-0.10	0.453	-0.22	0.833

S = 32.475 R-Sq = - % R-Sq(adj) = - %

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	-	33618	-	-	-
Residual Error	-	-	-	-	-
Total	-	44165	-	-	-

Source	DF	Seq SS
X1	1	15332
X2	1	9260
X3	1	8977
X4	1	50

Unusual Observations

Obs	X1	Y	Fit	SE Fit	Residual	St Resid
8	12.0	245.95	-	-	-	2.44R

R denotes an observation with a large standardized residual.

Predicted Values for New Observations

NewObs	Fit	SE Fit	95% CI	95% PI
1	153.03	9.21	(132.50, 173.56)	(77.82, 228.25)

Values of Predictors for New Observations

NewObs	X1	X2	X3	X4
1	10	12	25	60

[Total 35 marks]

Question 3

Complete parts (a) and (b).

- (a) The mineral compositions of a sample of 25 rock specimens of coxite type were recorded. Each composition consists of the percentage by weight of four minerals, Albite, Blandite, Cornite, and Daubite. The depth of location, and porosity of the sample were also recorded.

We will look for a model that explains porosity as a function of mineral composition and depth of location.

Output 1 includes a scatterplot matrix of the observations, along with the pairwise correlation coefficients.

Output 2 includes the fit for a model where all five explanatory variables are used, along with some diagnostic plots for that fit.

Output 3 includes output from the best subsets routine.

- (i) Write down the hypotheses, in terms of the model coefficients, β 's, being tested in the ANOVA table for this model.
Referencing the p-value printed for the ANOVA, what do you conclude as a result of this test:
- in terms of the model coefficients, β 's, and
 - in terms of the predictor variables? [5 marks]
- (ii) Using output 1 and output 2 to support your answer, discuss the predictive power of the five variables, by giving your conclusions from the outputs of the t-tests and ANOVA test provided. Ensure you provide the hypotheses being tested for any test that you reference. [5 marks]
- (iii) Carry out the partial F-test to test the hypothesis $H_0 : \beta_4 = \beta_5 = 0$, at $\alpha = 0.05$. Include a test statistic and rejection region. Interpret your findings in terms of the variables in this application. [5 marks]
- (iv) Explain the meaning of the phrase “a parsimonious model”. [2 marks]
- (v) Using output 3 to support your answer, write down the most suitable model, i.e. the set of predictors, that you think best explains the variability in observed **porosity**.
Justify your choice of model by explaining the role of s , R^2 , adjusted R^2 , and Mallow's C_p in choosing the best model. [8 marks]

- (b) A hospital surgical unit is interested in predicting survival in patients undergoing a particular type of liver operation. A random sample of 108 patients was selected for which a liver function test score was recorded prior to the operation. The survival times for these patients were observed in a follow up study.

Output 4 displays the results and diagnostic plots for a simple linear regression fit to the data. The output of the routine to fit the Box-Cox transformation is also included.

Question 3 continues on the next page

Question 3 continued...

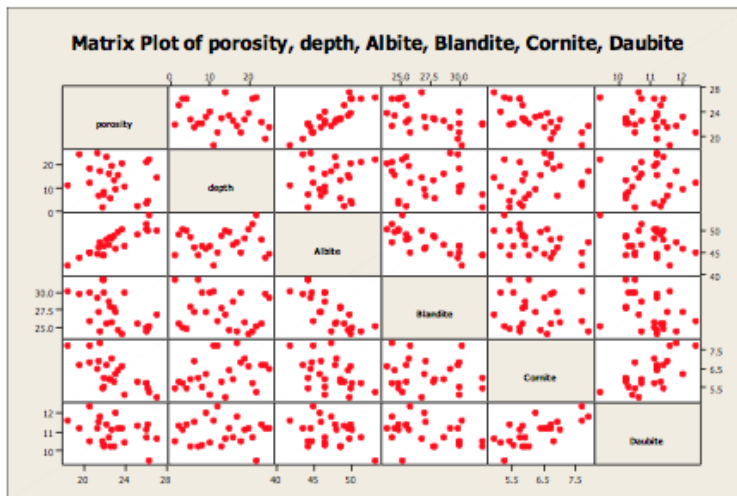
- (i) Comment on the validity of the assumptions for this regression fit. [4 marks]
- (ii) What does the output of the Box-Cox transformation indicate? [3 marks]
- (iii) A simple linear regression using *LiverTest* as the predictor was fitted to the transformed response variable $\ln(\text{SurvivalTime})$ giving the output 5.
Using this model predict the survival time of a patient whose pre-operation liver test score is observed to be 3. [3 marks]

Output 1:

Correlations: porosity, Albite, Blandite, Cornite, Daubite, depth

	porosity	Albite	Blandite	Cornite	Daubite
Albite	0.869				
Blandite	-0.551	-0.767			
Cornite	-0.723	-0.474	-0.114		
Daubite	-0.320	-0.314	-0.336	0.682	
depth	-0.147	0.091	-0.200	0.301	-0.016

Cell Contents: Pearson correlation



Question 3 continues on the next page

Question 3 continued...

Output 2:

The regression equation is

$$\text{porosity} = -218 + 0.0145 \text{ depth} + 2.65 \text{ Albite} + 2.19 \text{ Blandite} + 0.21 \text{ Cornite} + 4.95 \text{ Daubite}$$

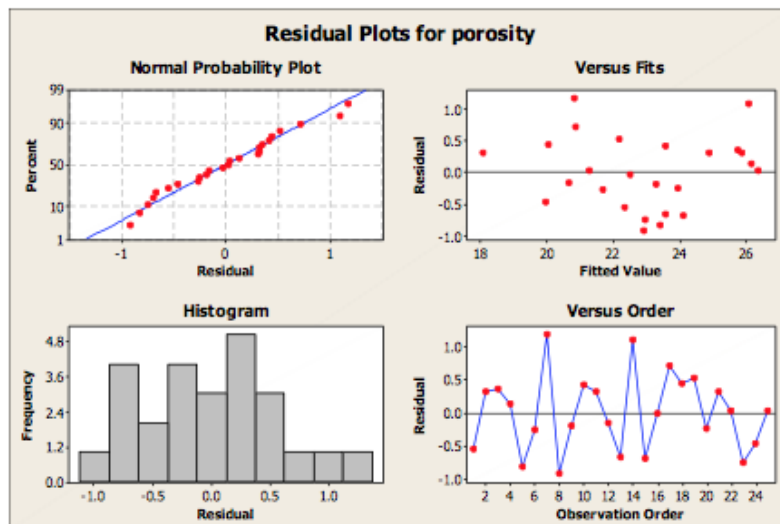
Predictor	Coef	SE Coef	T	P
Constant	-217.7	253.4	-0.86	0.401
depth	0.01448	0.03329	0.44	0.668
Albite	2.649	2.483	1.07	0.299
Blandite	2.191	2.601	0.84	0.410
Cornite	0.211	2.227	0.09	0.925
Daubite	4.949	4.672	1.06	0.303

S = 0.649388 R-Sq = 93.6% R-Sq(adj) = 91.9%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	5	116.248	23.250	55.13	0.000
Residual Error	19	8.012	0.422		
Total	24	124.260			

Source	DF	Seq SS
depth	1	2.678
Albite	1	97.563
Blandite	1	2.344
Cornite	1	13.189
Daubite	1	0.473



Question 3 continues on the next page

Question 3 continued...
Output 3:

Best Subsets Regression: porosity versus depth, Albite, ...

Response is porosity

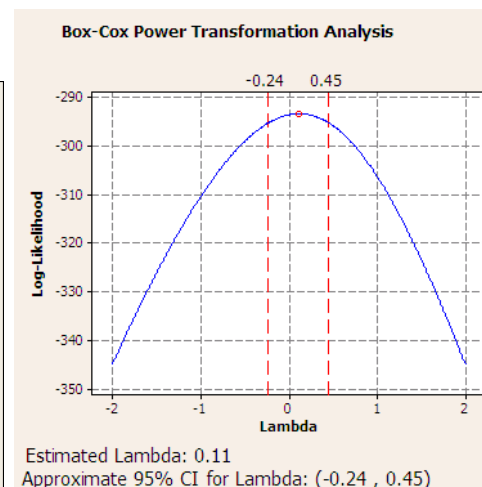
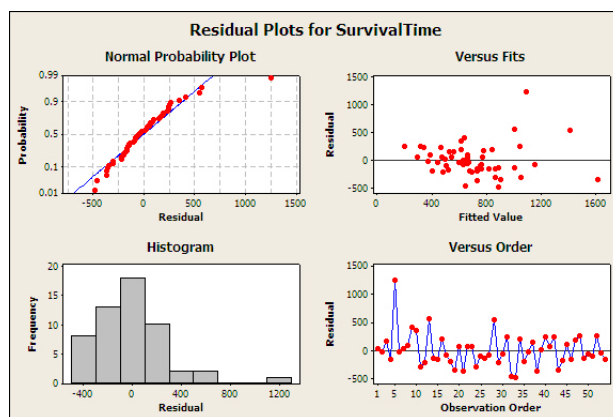
					B				
					l C D				
					A a o a				
					d l n r u				
					e b d n b				
					p i i i i				
					t t t t t				
					h e e e e				
Vars	R-Sq	R-Sq(adj)	Mallows Cp	S					
1	75.5	74.5	51.1	1.1500	X				
1	52.3	50.2	119.5	1.6050				X	
2	93.0	92.4	1.6	0.62779		X	X		
2	88.0	86.9	16.3	0.82215		X		X	
3	93.5	92.5	2.3	0.62254		X	X		X
3	93.3	92.3	2.8	0.63111		X		X	X
4	93.5	92.3	4.0	0.63309		X	X	X	X
4	93.5	92.2	4.2	0.63609		X	X	X	X
5	93.6	91.9	6.0	0.64939		X	X	X	X

Output 4:

The regression equation is
SurvivalTime = 15 + 250 LiverTest

Predictor	Coef	SE Coef	T	P
Constant	15.2	111.9	0.14	0.893
LiverTest	250.30	38.02	6.58	0.000

S = 296.299 R-Sq = 45.5% R-Sq(adj) = 44.4%



Question 3 continues on the next page

Question 3 continued...

Output 5:

The regression equation is

$\ln(\text{SurvivalTime}) = 5.61 + 0.298 \text{ LiverTest}$

Predictor	Coef	SE Coef	T	P
Constant	5.6122	0.1425	39.38	0.000
LiverTest	0.29819	0.04844	6.16	0.000

S = 0.377459 R-Sq = 42.2% R-Sq(adj) = 41.0%

[Total 35 marks]

Question 4

Complete parts (a), (b) and (c).

- (a) A researcher compared the inter-radial suture widths of urchins maintained on of three food regimes (Initial: no additional food supplied above what was in the initial sample, low: food supplied periodically and high: food supplied *ad libitum*).

Indicator variables are defined for the three food regimes:

$$\begin{aligned} IndInit : & \begin{cases} 1 & \text{if the urchin observed is maintained on the initial food regime} \\ 0 & \text{otherwise} \end{cases} \\ IndLow : & \begin{cases} 1 & \text{if the urchin observed is maintained on a low food regime} \\ 0 & \text{otherwise} \end{cases} \\ IndHigh : & \begin{cases} 1 & \text{if the urchin observed is maintained on a high food regime} \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

and the model, $SUTW_i = \beta_0 + \beta_1 IndHigh_i + \beta_2 IndInit_i + \epsilon_i$, is fitted to the data, giving:

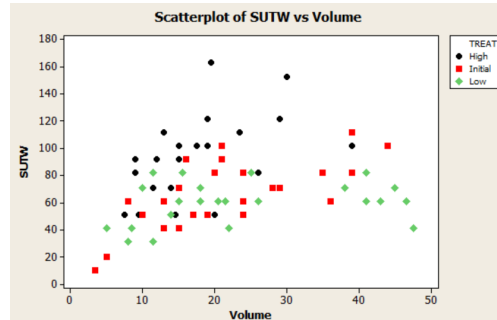
The regression equation is				
SUTW = 59.9 + 30.6 IndHigh + 6.79 IndInit				
Predictor	Coef	SE Coef	T	P
Constant	59.917	5.068	11.82	0.000
IndHigh	30.583	7.168	4.27	0.000
IndInit	6.792	7.168	0.95	0.347
S = 24.8301 R-Sq = 22.5% R-Sq(adj) = 20.3%				

- i) Use the fitted model to predict suture width for:
- an urchin on a high food regime.
 - an urchin on an initial food regime.
 - an urchin on a low food regime.
- [3 marks]
- ii) Interpret the meaning of the estimated coefficients of *IndHigh* and *IndInit*, and comment on their significance. Ensure you state the hypotheses being tested when referring to the results of the output.
- [6 marks]
- iii) Write down an alternative model proposal that you would recommend be fitted to the observed data that would help assess the comparison of suture width between each of the food regimes, initial and low, compared with a high food regime.
- [2 marks]

Question 4 continues on the next page

Question 4 continued...

- (b) In an attempt to control for substantial variability in urchin sizes, the initial body volume of each urchin was measured as a covariate. A plot of the observed relationship of SUTW with Volume is provided, with an indication to the different food regimes.



The regression analysis below models the response, *SUTW*, by the predictor *Volume*, and by the indicator variables *IndInit*, *IndLow*, as defined above.

The regression equation is

$$\text{SUTW} = 73.4 + 0.991 \text{ Volume} - 28.1 \text{ IndInit} - 37.0 \text{ IndLow}$$

Predictor	Coef	SE Coef	T	P
Constant	73.367	6.203	11.83	0.000
Volume	0.9908	0.2423	4.09	0.000
IndInit	-28.064	6.553	-4.28	0.000
IndLow	-36.982	6.655	-5.56	0.000

S = 22.4082 R-Sq = 37.8% R-Sq(adj) = 35.1%

- Interpret the meaning of the fitted coefficients of *Volume*, *IndInit* and *IndLow*, in this model. [5 marks]
 - For this model, write down the estimated relationship between SUTW and Volume for each of the three food regimes, Initial, Low and High. [3 marks]
 - What assumption is being made about the relationship of SUTW and Volume for the three different food regimes in this model? [2 marks]
- (c) Two interaction terms were added to the model:
Vol.IndInit, calculated as $(Volume_i) \times (IndInit_i)$ for each individual,
Vol.IndLow, calculated as $(Volume_i) \times (IndLow_i)$ for each individual,
giving the following output:

Question 4 continues on the next page

Question 4 continued...

The regression equation is

$$\text{SUTW} = 54.5 + 2.08 \text{ Volume} - 21.4 \text{ IndInit} - 1.6 \text{ IndLow} - 0.525 \text{ Vol.IndInit} - 1.78 \text{ Vol.IndLow}$$

Predictor	Coef	SE Coef	T	P
Constant	54.53	10.89	5.01	0.000
Volume	2.0800	0.5783	3.60	0.001
IndInit	-21.41	14.53	-1.47	0.145
IndLow	-1.63	13.96	-0.12	0.907
Vol.IndInit	-0.5254	0.7020	-0.75	0.457
Vol.IndLow	-1.7848	0.6607	-2.70	0.009

S = 21.1545 R-Sq = 46.2% R-Sq(adj) = 42.1%

- i) Interpret the meaning of the fitted coefficients of *Vol.IndInit* and *Vol.IndLow* in this model. [4 marks]
- ii) For this model, write down the estimated relationship between SUTW and Volume for each of the three food regimes, Initial, Low and High. [4 marks]
- iii) Determine if the terms *Vol.IndInit* and *Vol.IndLow* are significant, using the results of the individual t-tests in this analysis. Ensure you state the hypotheses being tested when referring to the results of the output. [6 marks]

[Total 35 marks]

ST311 Formulae

Some useful z-scores:

$$Z_{0.1} = 1.28$$

$$Z_{0.05} = 1.64$$

$$Z_{0.025} = 1.96$$

$$Z_{0.01} = 2.33$$

$$Z_{0.005} = 2.58$$

Some ranges for the standard normal distribution:

$$1.35 \quad \text{middle } 50\% : Z_{0.75} - Z_{0.25}$$

$$2.30 \quad \text{middle } 75\% : Z_{0.875} - Z_{0.125}$$

$$3.07 \quad \text{middle } 87.5\% : Z_{0.9375} - Z_{0.0625}$$

$$3.73 \quad \text{middle } 93.75\% : Z_{0.96875} - Z_{0.03125}$$

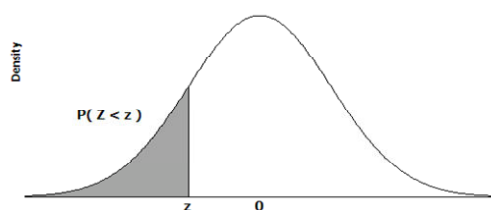
$$4.31 \quad \text{middle } 96.875\% : Z_{0.984375} - Z_{0.015625}$$

$$4.84 \quad \text{middle } 98.4375\% : Z_{0.9921875} - Z_{0.0078125}$$

Some formulae that may be useful when carrying out inference with parametric procedures:

- Inference for population correlation coefficient ρ : $TS = r\sqrt{\frac{n-2}{1-r^2}}$
- Partial F-test : $TS = \frac{(\text{additional SSreg})/q}{MS_{error}}$

Table: Cumulative Probabilities for the Standard Normal Distribution, $Z \sim \text{Normal}(0,1)$.



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Table: Cumulative Probabilities for the Standard Normal Distribution, $Z \sim \text{Normal}(0,1)$.

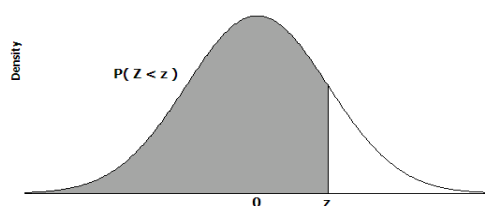
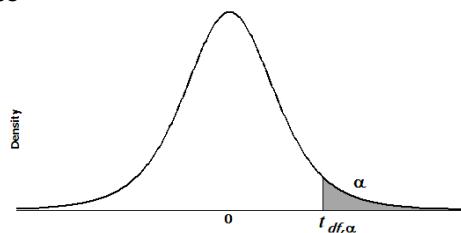
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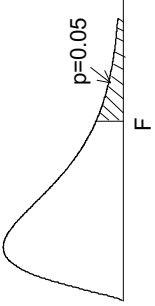
Table: t distribution critical values



df	Upper tail probability											
	0.25	0.20	0.15	0.10	0.05	0.025	0.02	0.01	0.005	0.0025	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.706	15.895	31.821	63.657	127.321	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.089	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
Z*	0.674	0.842	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.291

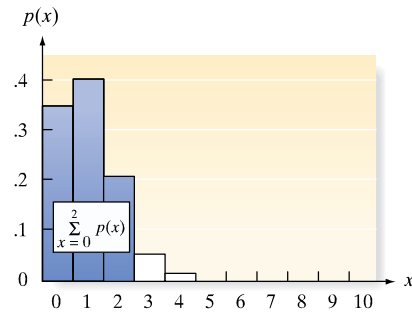
F distribution for $p = 0.05$

For various pairs of degrees of freedom (v_1, v_2), the tabled entries represent the critical values of F above which the proportion $p = 0.05$ of the distribution falls. (Example, for $df = 3, 8$ and $F = 2.92$ is exceeded by $p = 0.05$ of the population.



$p = 0.05$		Degrees of freedom for the numerator v_1																		
		v_2	1	2	3	4	5	6	7	8	9	10	12	15	20	30	40	60	120	∞
1	161.446	199.499	215.707	224.583	230.160	233.988	236.767	238.884	240.543	241.882	243.905	245.949	248.016	250.096	251.144	252.196	253.254	254.317		
2	18.513	19.000	19.164	19.247	19.296	19.329	19.353	19.371	19.385	19.396	19.412	19.429	19.446	19.463	19.471	19.479	19.487	19.496		
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.785	8.745	8.703	8.660	8.617	8.594	8.572	8.549	8.526		
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803	5.746	5.717	5.688	5.658	5.628		
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558	4.496	4.464	4.431	4.398	4.365		
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874	3.808	3.774	3.740	3.705	3.669		
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445	3.376	3.340	3.304	3.267	3.230		
8	5.318	4.459	4.066	3.838	3.688	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150	3.079	3.043	3.005	2.967	2.928		
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936	2.864	2.826	2.787	2.748	2.707		
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774	2.700	2.661	2.621	2.580	2.538		
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646	2.570	2.531	2.490	2.448	2.404		
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544	2.466	2.426	2.384	2.341	2.296		
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459	2.380	2.339	2.297	2.252	2.206		
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388	2.308	2.266	2.223	2.178	2.131		
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328	2.247	2.204	2.160	2.114	2.066		
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276	2.194	2.151	2.106	2.059	2.010		
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230	2.148	2.104	2.058	2.011	1.960		
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191	2.107	2.063	2.017	1.968	1.917		
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155	2.071	2.026	1.980	1.930	1.878		
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124	2.039	1.994	1.946	1.896	1.843		
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.176	2.096	2.010	1.965	1.916	1.866	1.812		
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.151	2.071	1.984	1.938	1.889	1.838	1.783		
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.128	2.048	1.961	1.914	1.865	1.813	1.757		
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	2.108	2.027	1.939	1.892	1.842	1.790	1.733		
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.092	2.015	1.932	1.841	1.792	1.740	1.683	1.622		
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077	2.003	1.924	1.839	1.744	1.693	1.637	1.577	1.509		
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.917	1.836	1.748	1.649	1.594	1.534	1.467	1.389		
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.910	1.834	1.750	1.659	1.554	1.495	1.429	1.352	1.254		
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.752	1.666	1.571	1.459	1.394	1.318	1.221	1.000		

TABLE II Binomial Probabilities



Tabulated values are $\sum_{x=0}^k p(x)$. (Computations are rounded at the third decimal place.)

a. $n = 5$

$\begin{matrix} p \\ k \end{matrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.951	.774	.590	.328	.168	.078	.031	.010	.002	.000	.000	.000	.000
1	.999	.977	.919	.737	.528	.337	.188	.087	.031	.007	.000	.000	.000
2	1.000	.999	.991	.942	.837	.683	.500	.317	.163	.058	.009	.001	.000
3	1.000	1.000	1.000	.993	.969	.913	.812	.663	.472	.263	.081	.023	.001
4	1.000	1.000	1.000	1.000	.998	.990	.969	.922	.832	.672	.410	.226	.049

b. $n = 6$

$\begin{matrix} p \\ k \end{matrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.941	.735	.531	.262	.118	.047	.016	.004	.001	.000	.000	.000	.000
1	.999	.967	.886	.655	.420	.233	.109	.041	.011	.002	.000	.000	.000
2	1.000	.998	.984	.901	.744	.544	.344	.179	.070	.017	.001	.000	.000
3	1.000	1.000	.999	.983	.930	.821	.656	.456	.256	.099	.016	.002	.000
4	1.000	1.000	1.000	.998	.989	.959	.891	.767	.580	.345	.114	.033	.001
5	1.000	1.000	1.000	1.000	.999	.996	.984	.953	.882	.738	.469	.265	.059

c. $n = 7$

$\begin{matrix} p \\ k \end{matrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.932	.698	.478	.210	.082	.028	.008	.002	.000	.000	.000	.000	.000
1	.998	.956	.850	.577	.329	.159	.063	.019	.004	.000	.000	.000	.000
2	1.000	.996	.974	.852	.647	.420	.227	.096	.029	.005	.000	.000	.000
3	1.000	1.000	.997	.967	.874	.710	.500	.290	.126	.033	.003	.000	.000
4	1.000	1.000	1.000	.995	.971	.904	.773	.580	.353	.148	.026	.004	.000
5	1.000	1.000	1.000	1.000	.996	.981	.937	.841	.671	.423	.150	.044	.002
6	1.000	1.000	1.000	1.000	1.000	.998	.992	.972	.918	.790	.522	.302	.068

(continued)

TABLE II (continued)

d. $n = 8$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.923	.663	.430	.168	.058	.017	.004	.001	.000	.000	.000	.000	.000
1	.997	.943	.813	.503	.255	.106	.035	.009	.001	.000	.000	.000	.000
2	1.000	.994	.962	.797	.552	.315	.145	.050	.011	.001	.000	.000	.000
3	1.000	1.000	.995	.944	.806	.594	.363	.174	.058	.010	.000	.000	.000
4	1.000	1.000	1.000	.990	.942	.826	.637	.406	.194	.056	.005	.000	.000
5	1.000	1.000	1.000	.999	.989	.950	.855	.685	.448	.203	.038	.006	.000
6	1.000	1.000	1.000	1.000	.999	.991	.965	.894	.745	.497	.187	.057	.003
7	1.000	1.000	1.000	1.000	1.000	.999	.996	.983	.942	.832	.570	.337	.077

e. $n = 9$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.914	.630	.387	.134	.040	.010	.002	.000	.000	.000	.000	.000	.000
1	.997	.929	.775	.436	.196	.071	.020	.004	.000	.000	.000	.000	.000
2	1.000	.992	.947	.738	.463	.232	.090	.025	.004	.000	.000	.000	.000
3	1.000	.999	.992	.914	.730	.483	.254	.099	.025	.003	.000	.000	.000
4	1.000	1.000	.999	.980	.901	.733	.500	.267	.099	.020	.001	.000	.000
5	1.000	1.000	1.000	.997	.975	.901	.746	.517	.270	.086	.008	.001	.000
6	1.000	1.000	1.000	1.000	.996	.975	.910	.768	.537	.262	.053	.008	.000
7	1.000	1.000	1.000	1.000	1.000	.996	.980	.929	.804	.564	.225	.071	.003
8	1.000	1.000	1.000	1.000	1.000	1.000	.998	.990	.960	.866	.613	.370	.086

f. $n = 10$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.904	.599	.349	.107	.028	.006	.001	.000	.000	.000	.000	.000	.000
1	.996	.914	.736	.376	.149	.046	.011	.002	.000	.000	.000	.000	.000
2	1.000	.988	.930	.678	.383	.167	.055	.012	.002	.000	.000	.000	.000
3	1.000	.999	.987	.879	.650	.382	.172	.055	.011	.001	.000	.000	.000
4	1.000	1.000	.998	.967	.850	.633	.377	.166	.047	.006	.000	.000	.000
5	1.000	1.000	1.000	.994	.953	.834	.623	.367	.150	.033	.002	.000	.000
6	1.000	1.000	1.000	.999	.989	.945	.828	.618	.350	.121	.013	.001	.000
7	1.000	1.000	1.000	1.000	.998	.988	.945	.833	.617	.322	.070	.012	.000
8	1.000	1.000	1.000	1.000	1.000	.998	.989	.954	.851	.624	.264	.086	.004
9	1.000	1.000	1.000	1.000	1.000	1.000	.999	.994	.972	.893	.651	.401	.096

(continued)

TABLE II (continued)

g. $n = 15$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.860	.463	.206	.035	.005	.000	.000	.000	.000	.000	.000	.000	.000
1	.990	.829	.549	.167	.035	.005	.000	.000	.000	.000	.000	.000	.000
2	1.000	.964	.816	.398	.127	.027	.004	.000	.000	.000	.000	.000	.000
3	1.000	.995	.944	.648	.297	.091	.018	.002	.000	.000	.000	.000	.000
4	1.000	.999	.987	.838	.515	.217	.059	.009	.001	.000	.000	.000	.000
5	1.000	1.000	.998	.939	.722	.403	.151	.034	.004	.000	.000	.000	.000
6	1.000	1.000	1.000	.982	.869	.610	.304	.095	.015	.001	.000	.000	.000
7	1.000	1.000	1.000	.996	.950	.787	.500	.213	.050	.004	.000	.000	.000
8	1.000	1.000	1.000	.999	.985	.905	.696	.390	.131	.018	.000	.000	.000
9	1.000	1.000	1.000	1.000	.996	.966	.849	.597	.278	.061	.002	.000	.000
10	1.000	1.000	1.000	1.000	.999	.991	.941	.783	.485	.164	.013	.001	.000
11	1.000	1.000	1.000	1.000	1.000	.998	.982	.909	.703	.352	.056	.005	.000
12	1.000	1.000	1.000	1.000	1.000	1.000	.996	.973	.873	.602	.184	.036	.000
13	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.995	.965	.833	.451	.171	.010
14	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.995	.965	.794	.537	.140

h. $n = 20$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.818	.358	.122	.012	.001	.000	.000	.000	.000	.000	.000	.000	.000
1	.983	.736	.392	.069	.008	.001	.000	.000	.000	.000	.000	.000	.000
2	.999	.925	.677	.206	.035	.004	.000	.000	.000	.000	.000	.000	.000
3	1.000	.984	.867	.411	.107	.016	.001	.000	.000	.000	.000	.000	.000
4	1.000	.997	.957	.630	.238	.051	.006	.000	.000	.000	.000	.000	.000
5	1.000	1.000	.989	.804	.416	.126	.021	.002	.000	.000	.000	.000	.000
6	1.000	1.000	.998	.913	.608	.250	.058	.006	.000	.000	.000	.000	.000
7	1.000	1.000	1.000	.968	.772	.416	.132	.021	.001	.000	.000	.000	.000
8	1.000	1.000	1.000	.990	.887	.596	.252	.057	.005	.000	.000	.000	.000
9	1.000	1.000	1.000	.997	.952	.755	.412	.128	.017	.001	.000	.000	.000
10	1.000	1.000	1.000	.999	.983	.872	.588	.245	.048	.003	.000	.000	.000
11	1.000	1.000	1.000	1.000	.995	.943	.748	.404	.113	.010	.000	.000	.000
12	1.000	1.000	1.000	1.000	.999	.979	.868	.584	.228	.032	.000	.000	.000
13	1.000	1.000	1.000	1.000	1.000	.994	.942	.750	.392	.087	.002	.000	.000
14	1.000	1.000	1.000	1.000	1.000	.998	.979	.874	.584	.196	.011	.000	.000
15	1.000	1.000	1.000	1.000	1.000	1.000	.994	.949	.762	.370	.043	.003	.000
16	1.000	1.000	1.000	1.000	1.000	1.000	.999	.984	.893	.589	.133	.016	.000
17	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.996	.965	.794	.323	.075	.001
18	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.992	.931	.608	.264	.017
19	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.988	.878	.642	.182

(continued)

TABLE XIV Critical Values of T_L and T_U for the Wilcoxon Rank Sum Test: Independent Samples

Test statistic is the rank sum associated with the smaller sample (if equal sample sizes, either rank sum can be used).

a. $\alpha = .025$ one-tailed; $\alpha = .05$ two-tailed

$n_2 \backslash n_1$	3		4		5		6		7		8		9		10	
	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U
3	5	16	6	18	6	21	7	23	7	26	8	28	8	31	9	33
4	6	18	11	25	12	28	12	32	13	35	14	38	15	41	16	44
5	6	21	12	28	18	37	19	41	20	45	21	49	22	53	24	56
6	7	23	12	32	19	41	26	52	28	56	29	61	31	65	32	70
7	7	26	13	35	20	45	28	56	37	68	39	73	41	78	43	83
8	8	28	14	38	21	49	29	61	39	73	49	87	51	93	54	98
9	8	31	15	41	22	53	31	65	41	78	51	93	63	108	66	114
10	9	33	16	44	24	56	32	70	43	83	54	98	66	114	79	131

b. $\alpha = .05$ one-tailed; $\alpha = .10$ two-tailed

$n_2 \backslash n_1$	3		4		5		6		7		8		9		10	
	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U
3	6	15	7	17	7	20	8	22	9	24	9	27	10	29	11	31
4	7	17	12	24	13	27	14	30	15	33	16	36	17	39	18	42
5	7	20	13	27	19	36	20	40	22	43	24	46	25	50	26	54
6	8	22	14	30	20	40	28	50	30	54	32	58	33	63	35	67
7	9	24	15	33	22	43	30	54	39	66	41	71	43	76	46	80
8	9	27	16	36	24	46	32	58	41	71	52	84	54	90	57	95
9	10	29	17	39	25	50	33	63	43	76	54	90	66	105	69	111
10	11	31	18	42	26	54	35	67	46	80	57	95	69	111	83	127

Source: From F. Wilcoxon, and R. A. Wilcox. "Some Rapid Approximate Statistical Procedures," 1964, 20–23. Courtesy of Lederle Laboratories Division of American Cyanamid Company, Madison, NJ.